4 **(a)** With reference to the vibrations of the particles on a wave and its direction of energy transfer, distinguish between transverse waves and longitudinal waves.

.....

.....

.....

..... [2]

(b) Fig. 4.1 shows the equilibrium positions of the particles in a medium when they are undisturbed as compared to their displaced positions in the presence of a progressive sinusoidal wave. Displacement to the right is taken as positive.

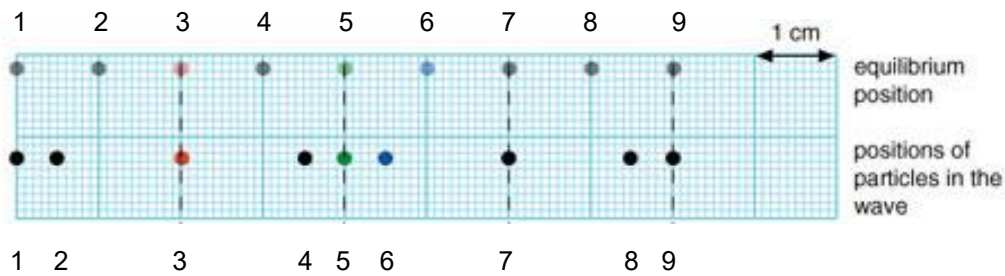

Fig. 4.1

(i) Deduce the displacement of particle 2 when the wave passes through it at the instant shown in Fig. 4.1.

displacement = ... cm [1]

(ii) Draw, using the axes provided below, the displacement-position graph representing all the particles disturbed by the wave as shown in Fig. 4.1. Indicate on your graph all centers of rarefaction R and compression C.

[3]

(iii) Hence deduce the amplitude of the progressive wave.

amplitude = ……. cm [1]

(iv) State the phase difference between particles 5 and 7.